Auto Grade Using Natural Language Processing

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CONTENTS

ABSTRACT

This project is designed to create a smart system that can generate and evaluate both subjective and objective answers using Natural Language Processing (NLP). It starts by loading the required libraries and datasets. Users can then choose whether they want to work with subjective or objective evaluations. For subjective answers, the system asks questions and allows users to type their responses. For objective tasks, the evaluation is done automatically. The system uses NLP techniques to check and validate the answers, and then records the results in an Excel sheet.

Overall, this solution makes the evaluation process faster and more accurate. By automating both the generation and validation of answers, it helps researchers and analysts handle large amounts of text data more efficiently and gain valuable insights with ease.

INTRODUCTION

As Natural Language Processing (NLP) continues to grow rapidly, there's a rising need for systems that can not only generate questions but also evaluate subjective answers accurately. This project offers a complete solution by using advanced NLP techniques to simplify and improve the process of assessing written responses. It combines powerful text-processing tools with a user-friendly interface, making it a practical and efficient tool for researchers and analysts. The system follows a well-structured workflow that supports both subjective and objective evaluations. It all begins by importing the necessary libraries and datasets. In this stage, the text is carefully cleaned and relevant questions are generated using the Natural Language Toolkit (NLTK).

OBJECTIVES

- **Develop a robust NLP system** for efficiently generating and evaluating **subjective and objective answers**.
- Integrate **cutting-edge text processing techniques** to ensure accurate and streamlined evaluation.
- Provide **a user-friendly interface** for researchers and analysts, powered by **advanced NLP algorithms**.
- Enable **automated text cleaning** and support for **dual assessment modes** (subjective and objective).
- Facilitate **systematic storage and analysis** of evaluation results for deeper insights and continuous improvement of NLP evaluation systems.

EXISTING SYSTEM

- **Rising demand** for efficient evaluation of textual data presents challenges in **Natural Language Processing (NLP)**.
- **Existing methods** often lack comprehensive solutions for handling both **subjective and objective assessments**.
- **Manual evaluation** introduces inefficiencies, lowers reliability, and reduces productivity and accuracy.
- The project aims to develop a **comprehensive NLP-driven evaluation system** to address these gaps.
- By combining **advanced text processing algorithms** with a **user-friendly interface**, the system improves workflow efficiency, **enhances evaluation accuracy**, and supports robust textual data analysis.

PROPOSED SYSTEM

•**Automates validation**, improves accuracy, and helps researchers derive **fast and reliable insights** from textual content.

•**Utilizes advanced NLP techniques** to efficiently evaluate both subjective and objective textual data.

•**Workflow begins** with importing necessary libraries and datasets.

•**Text is cleaned** and **questions are generated** using the NLTK toolkit.

•Users can choose between:

•**Subjective evaluation** – users respond to predefined questions.

•**Objective evaluation** – responses are assessed automatically.

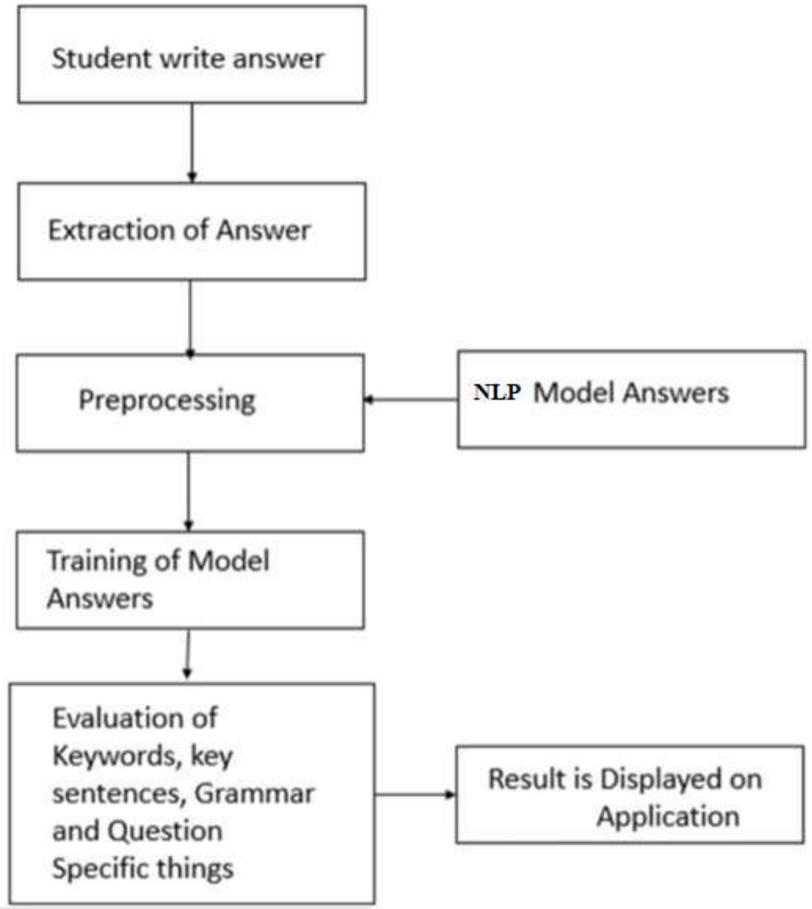
•**NLP algorithms** are applied to **validate and evaluate** responses with high accuracy.

•**Results are stored in Excel** files for easy access and future reference.

LITERATURE SURVEY

Literature	Techniques Used	Limitations	Proposed Solutions
Pang & Lee (2008)	Sentiment analysis using Naive Bayes, SVM	Limited to binary classification (positive/negative). Doesn’t capture context or deeper emotions.	Use BERT-based models to understand a wider range of emotions, including sarcasm and irony.
Liu et al. (2017)	Deep learning models for grammar correction	Misses complex grammatical errors and struggles with domain-specific language.	Use transformer for better grammar and style checks in different fields.
Devlin et al. (2018)	BERT-based models for factual correctness and answer validation	Needs large datasets and struggles with facts in specific areas or unclear information.	Fine-tune BERT models for specific topics to improve accuracy when checking facts.
Hutto & Gilbert (2014)	VADER for sentiment analysis; NLP models for evaluation	NLP models often show bias, like gender or racial biases, affecting the fairness of results.	Use bias-reducing techniques like counterfactual fairness to make the evaluation.
Reimers & Gurevych (2019)	Sentence-BERT for semantic similarity	NLP models can be slow and computationally heavy, making them hard to scale for large datasets.	Optimize models for faster performance and better handling of large amounts of data, making them quicker and more efficient.

SYSTEM ARCHITECTURE



REQUIREMENTS

- SOFTWARE
  - 1) Software : Anaconda
  - 2) Primary Language : Python
  - 3) Frontend Framework : Flask
  - 4) Back-end Framework : Jupyter Notebook
  - 5) Database : Sqlite3
  - 6) Front-End Technologies : HTML, CSS, JavaScript and Bootstrap4

REQUIREMENTS

- HARDWARE
  - 1) Operating System : Windows Only
  - 2) Processor : i5 and above
  - 3) Ram : 8gb and above
  - 4) Hard Disk : 25 GB in local drive

CONCLUSION

•This project marks a **significant milestone in NLP**, addressing the need for robust and efficient evaluation systems.

•Seamlessly integrates **advanced NLP techniques** with a **user-friendly interface** to streamline textual assessment for researchers and analysts.

•Implements a **systematic and efficient workflow** for both subjective and objective evaluations using **meticulous text processing**.

•Ensures **accuracy and reliability** in assessments through **sophisticated NLP algorithms**, increasing trust in results.

•**Systematic storage** of results allows for easy access and supports **further analysis** and **scalability**.

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THANK YOU